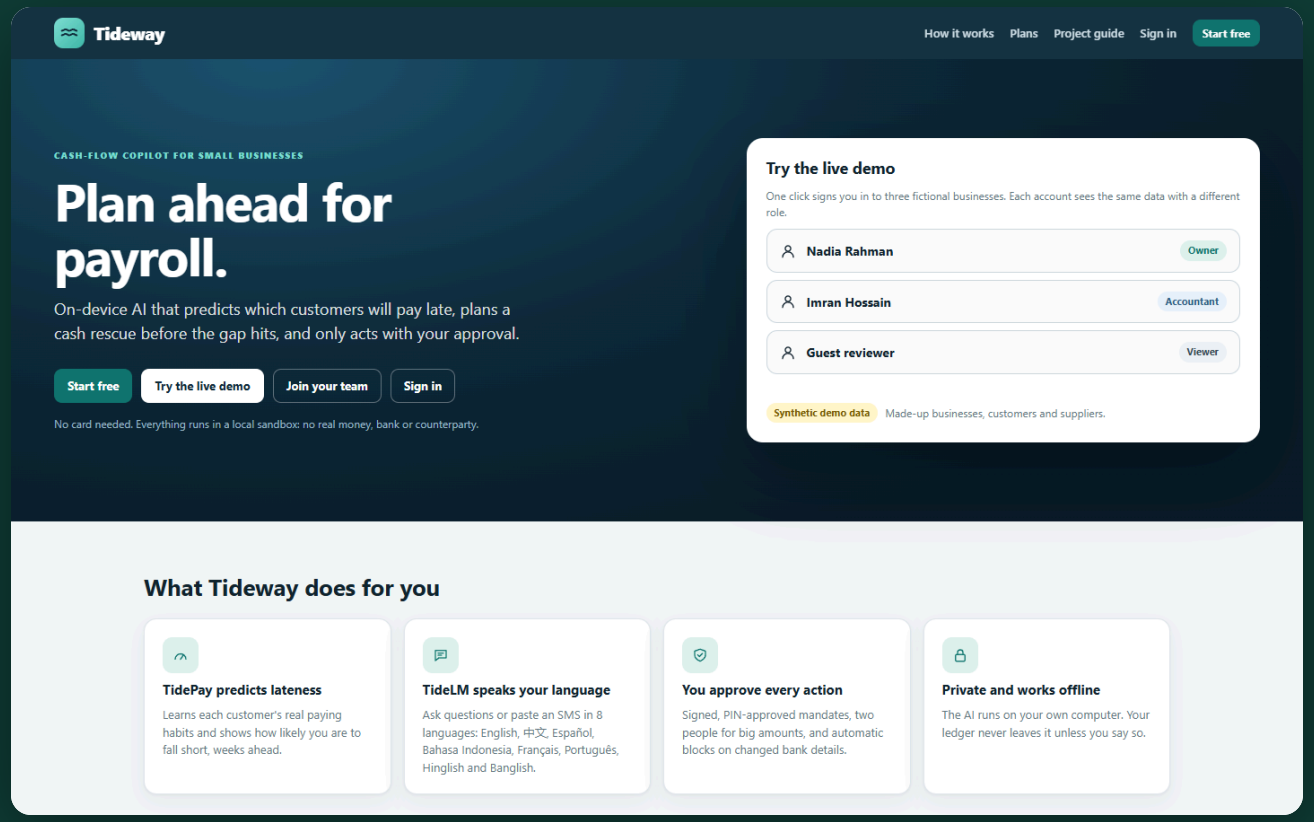
TIDEWAY · TOPIC E

# Tideway

## Plan ahead for payroll.

- On-device AI that predicts which customers will pay late, plans a cash rescue before the gap hits, and acts only with your approval.
- FinTechathon 2026 · International Track · Topic E: SME Finance Copilot
- The Last BrainCell



# The moment that matters

Payroll is days away and the cash is not there

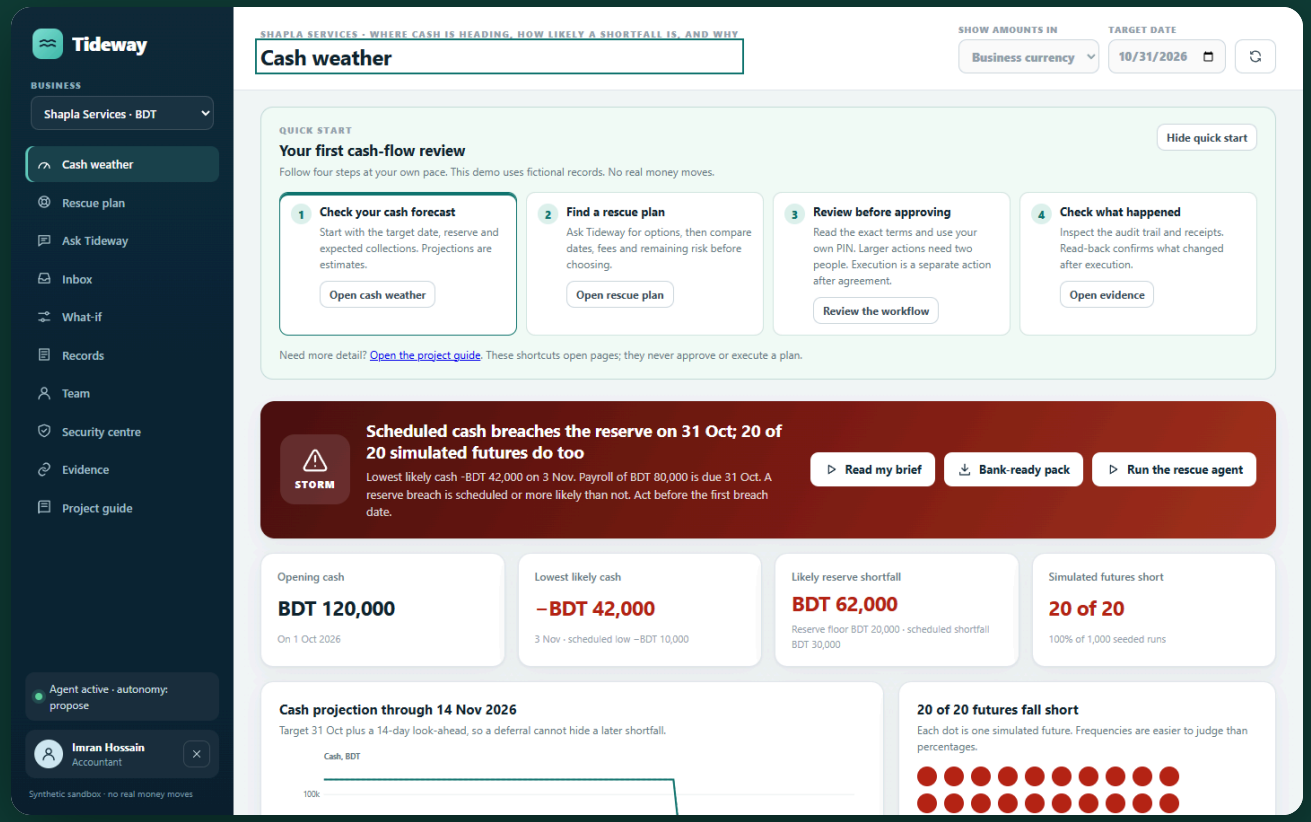
- Small businesses are paid late and pay on time: the gap lands on payroll day.
- Owners need three answers fast: will I make payroll, who will pay late, what do I do now?
- And an AI that helps must never be able to move money on its own.

TIDEWAY · TOPIC E

# See it coming

Cash weather, three scenarios, 1,000 simulated futures

- Exact daily ledger in minor units with a 14-day look-ahead
- Scheduled, likely and stress views; "N of 20 futures fall short"
- Shapla Services (synthetic): scheduled low −BDT 10,000 on 31 Oct; reserve gap BDT 30,000



# TidePay: chronological audit

Same held-out invoices, training-only statistics

| Model | MAE days | Log loss | Brier sum |
|---|---|---|---|
| Prior | 11.986 | 1.588 | **0.794** |
| Heuristic | 7.411 | 1.569 | **0.790** |
| Local-only | 7.215 | 1.233 | **0.641** |
| Federated | 6.957 | 1.073 | **0.578** |
| Pooled | 6.945 | 1.093 | **0.586** |

- 2376 test invoices, 40 synthetic SMEs; historical prior leakage disclosed
- Audit benchmark, shipped weights unchanged; pooled MAE slightly better
- One-process federation with pooled scaling; no deployed privacy guarantee

# TideLM: our own on-device language model

Qwen3-4B + a LoRA adapter we trained on a laptop with MLX

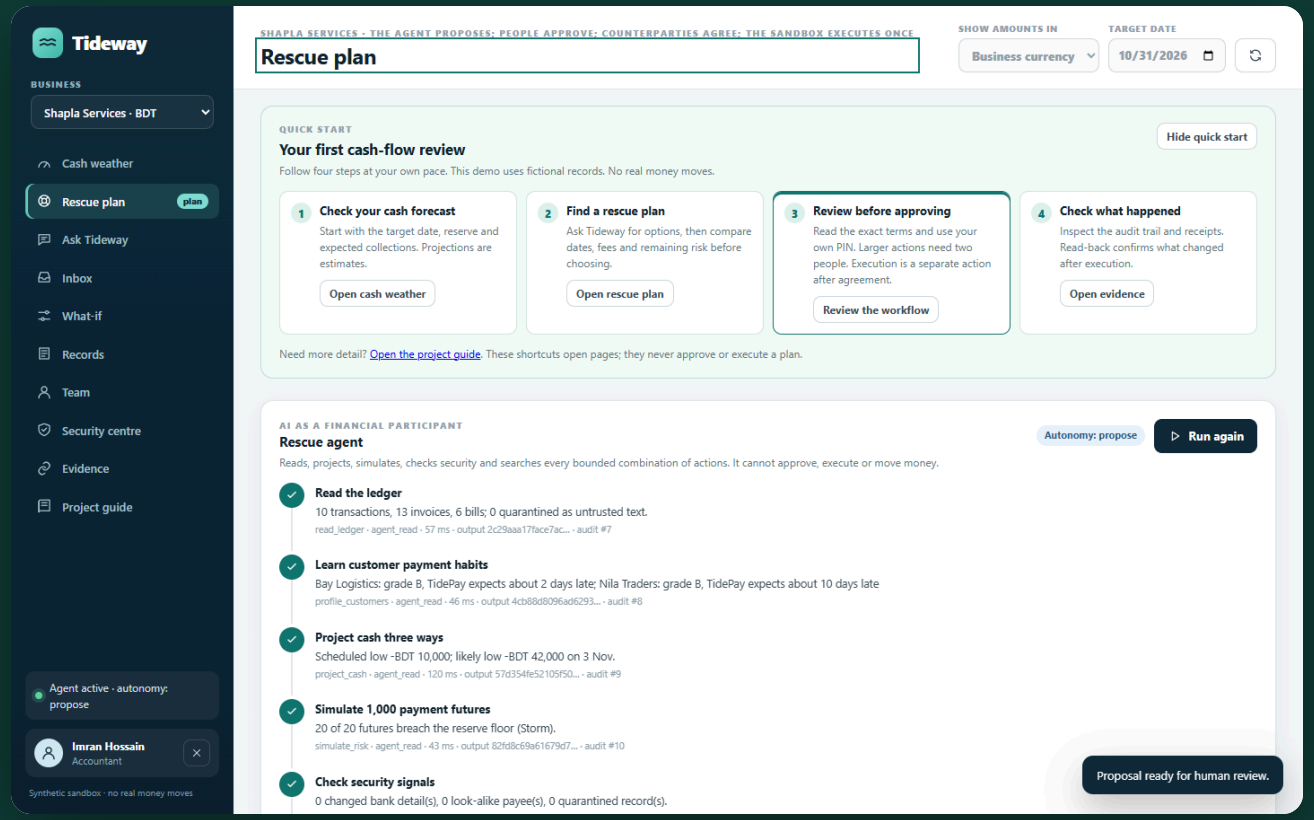
| | Base model | TideLM |
|---|---|---|
| Exact plan (every tool and argument) | 26.7% | **96.7%** |
| Understands the request (intent) | 20% | **99%** |
| Reads records from messages (F1) | 0.834 | **0.969** |
| Flags hidden instructions | 91.5% | **98.5%** |

- Eight languages: English, 中文, Español, Bahasa, Français, Português, Hinglish, Banglish
- Synthetic test with unseen names and shared template families
- Windows compatible model: 6/6 authored smoke requests matched; historical adapter scores above are a different evaluation

# Plan the rescue

An exact, risk-aware optimiser

- Supplier deferrals, early-payment discounts (priced as APR), zero-cash reminders
- Feasible plans protect payroll and the likely reserve path; residual risk stays visible
- Explains why the cheaper option is too risky; financing shown as a mock comparison

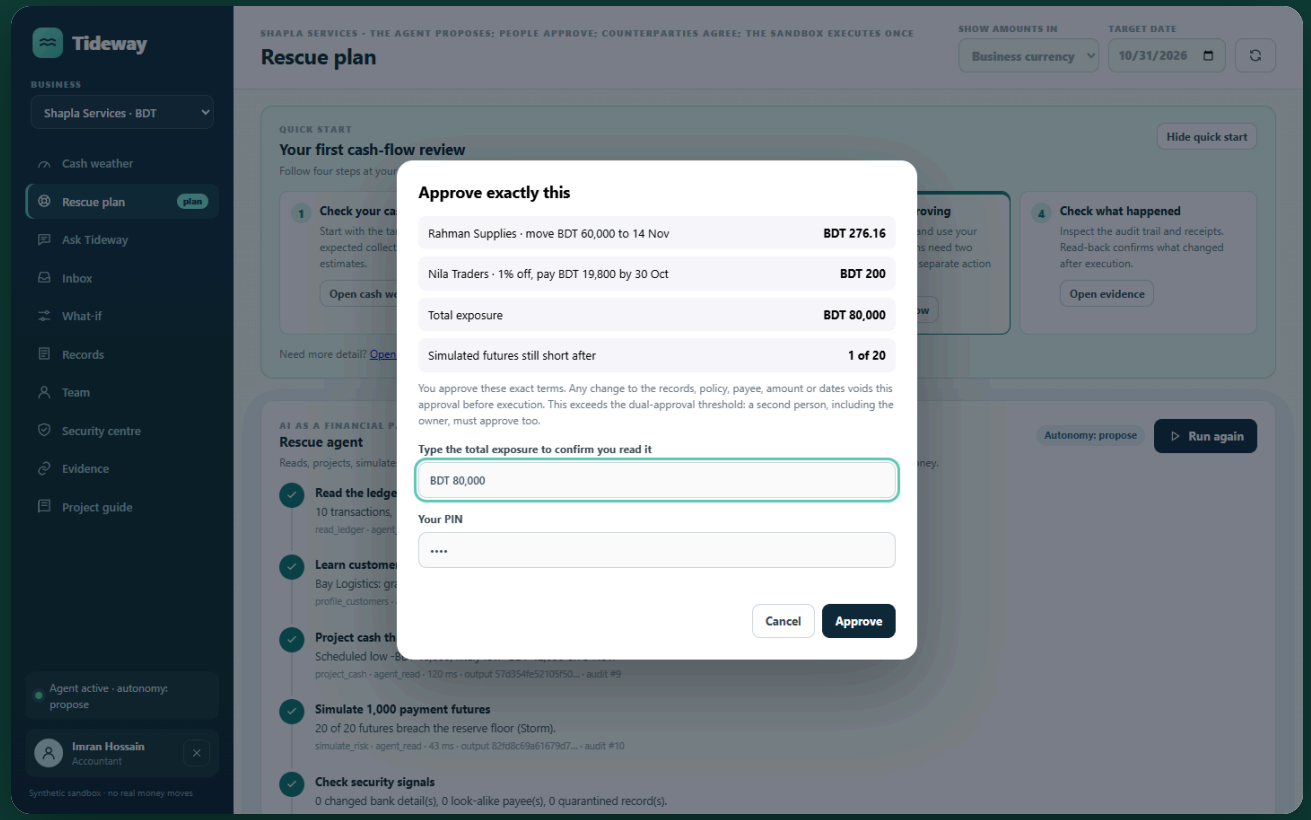


TIDEWAY · TOPIC E

# Approve exactly this

Human approval, signed mandate, verified execution

- Maker-checker with PIN; two people and a typed amount above the threshold
- HMAC-signed mandate bound to actions, amounts, snapshot and expiry (AP2-style)
- Simulated counterparties accept, counter or refuse; idempotent execution with read-back



# A delay changes the decision

Apply a real synthetic-ledger change; old authority is void

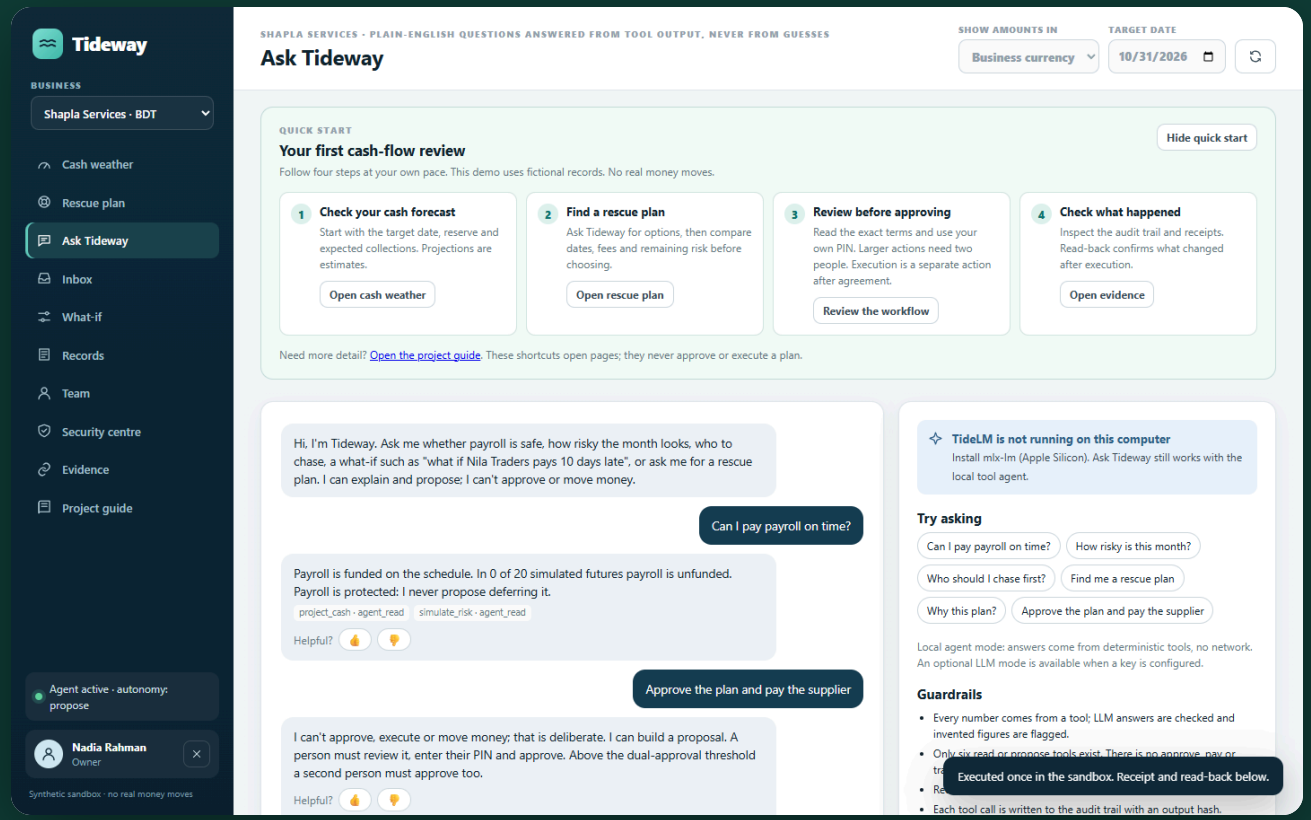
- Bay Logistics: 2026-11-03 to 2026-11-15
- Likely minimum BDT -42,000 to -57,000
- Old mandate refused; new proposal replaces supplier deferral with early-payment offers
- Bay counters at 1.5%; fresh approvals, consent, execute once and verify
- Runbook: docs/PAYMENT_DELAY_RUNBOOK.md; no video recorded

TIDEWAY · TOPIC E

# Talk to it in your language

Glass-box answers, zero-entry bookkeeping

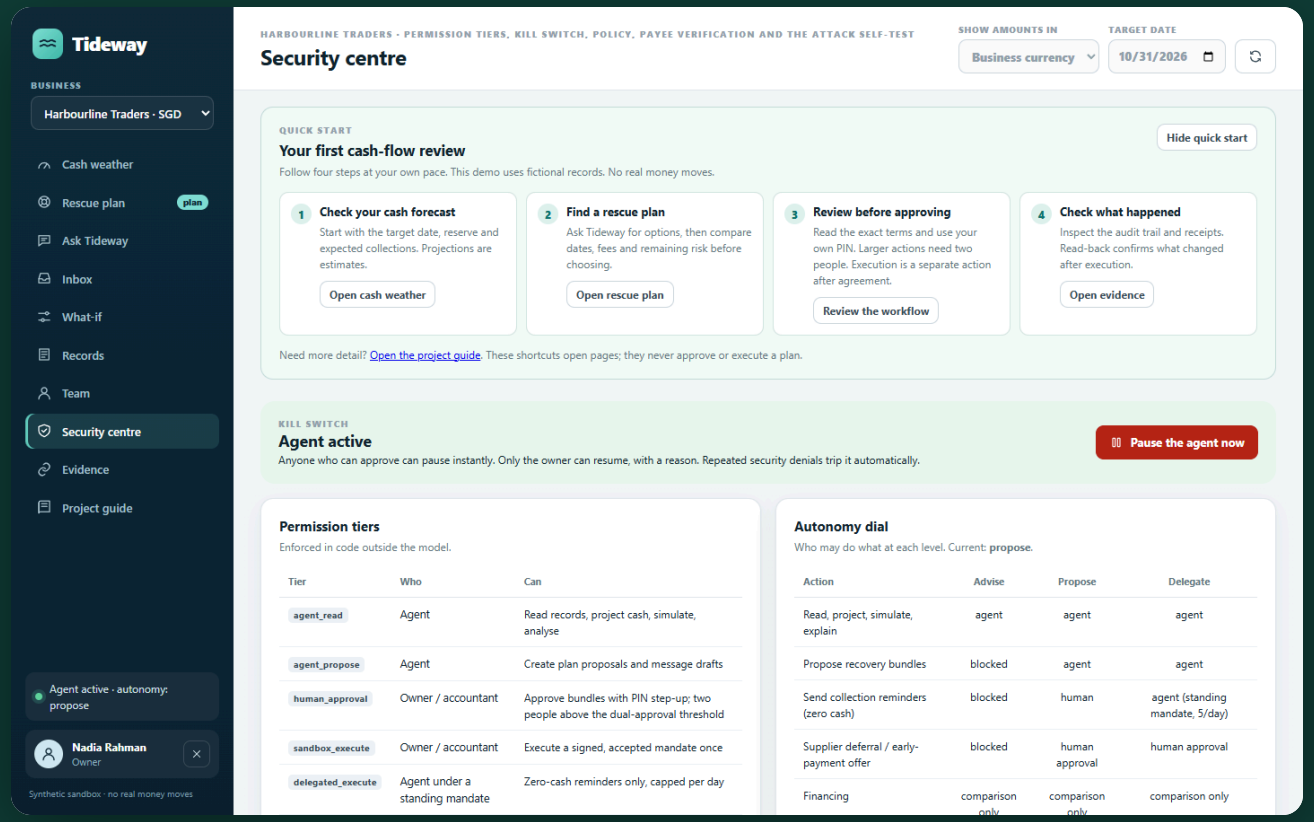
- Optional TideLM shows its typed plan; figures come from Tideway's tools
- Answers in the language of the question, with the same figures
- Optional local models draft messages and photos; human confirmation is required



# Security enforcement

Permissions and validation live outside the model

- 25/25 attacks blocked, 5/5 tasks completed, assuming a hijacked model
- Injection and payee-fraud screening in eight languages; changed accounts and look-alikes blocked
- Kill switch, circuit breaker, separation of duties, hash-chained audit with witness

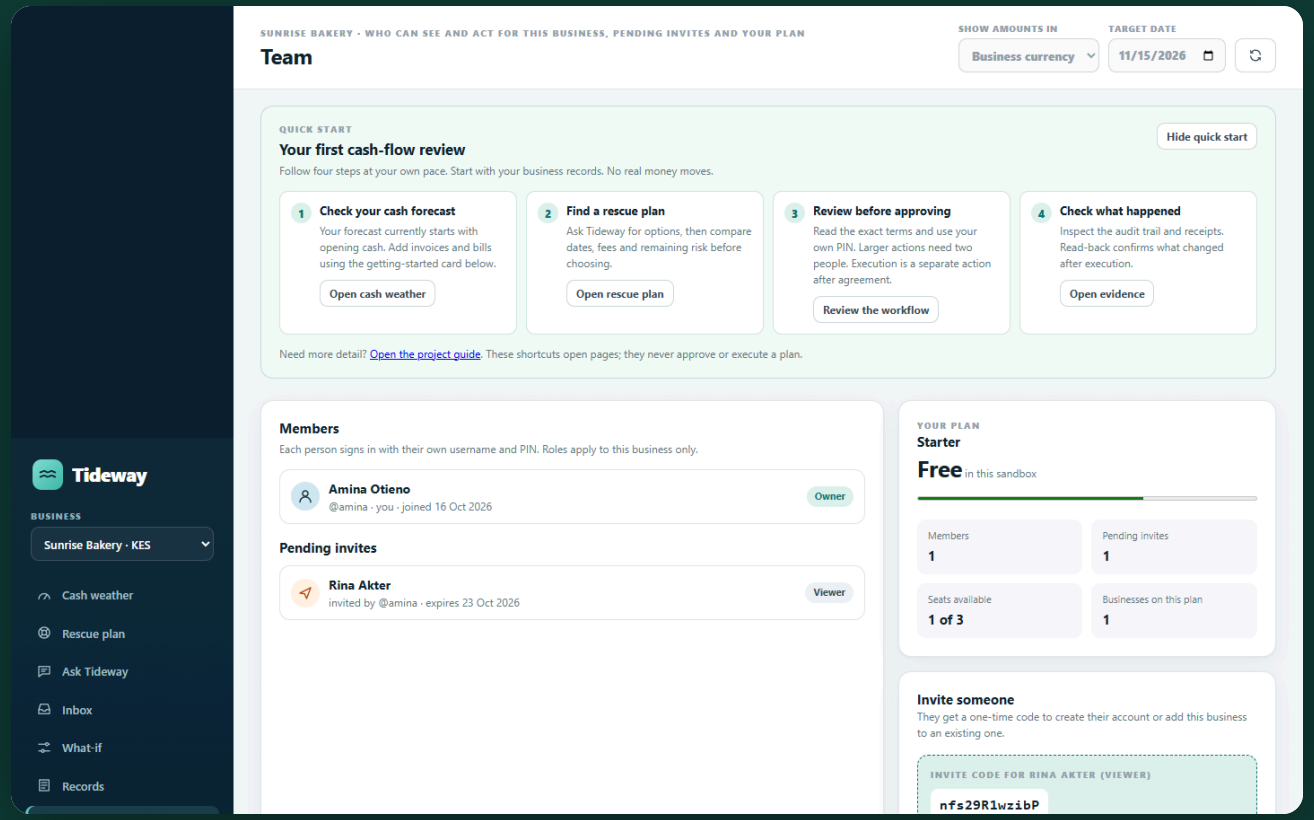


TIDEWAY · TOPIC E

# Built as a product

## Workspaces, teams and roles

- Sign up a business in any of 29 currencies; demo in Bangladesh, China and Singapore
- Invite your accountant by one-time code; roles per business; tenants never see each other
- Counterparty risk map: who can hurt your cash, and by how much



# Beyond three demo fixtures

24 distinct conditions; development and held-out expectations sealed

- 24/24 outcome handling: 9 rescues, 9 infeasible, 2 no-action, 4 blocked
- Only 7/9 rescues meet the 10% simulated-risk target
- Injection: 12/14 attacks detected; 7/14 benign messages flagged
- Payee: 6/8 attacks detected; 3/8 benign cases flagged
- Retained misses, raw outcomes and planner timings: evidence/upgrade-v1/scenarios/

# Evidence, with denominators

Reproducible in one command each

- 30/30 self-test cases · 198 automated tests
- Model reports, sandbox logs, browser evidence and reproducible setup
- python3 -m tideway.selftest · python3 -m unittest discover -s tests

# Honest limits, clear next steps

What we have not done yet

- Synthetic data and simulated counterparties; demo-grade PINs and keys
- No user study or external penetration test yet
- Next: consented SME pilots, a real bank rail behind the same mandate, federated TidePay across real users

TIDEWAY · TOPIC E

# The Last BrainCell

Bangladesh University of Engineering and Technology

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- Shaharia Rahman Santo
- Md. Atik Khan
- Md. Ashraf Hossain Chowdhury
- Khandaker Shadman Sakib Siddiquee